

Feature A

Known: $W = 600 \text{ lbf}$ $l = 1.0 \text{ in}$ $E = 10 \times 10^6 \text{ psi}$ $\delta = 0.005 \text{ in}$
 Unknown: Minimum radius r

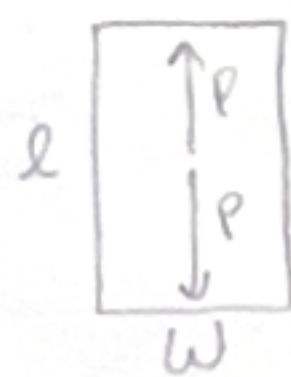


$$\delta = \frac{W \cdot l^3}{8 \cdot E \cdot I} \quad I = \frac{\pi r^4}{4} \quad r = \sqrt[4]{\frac{W \cdot l^3}{2\pi \cdot E \cdot \delta}}$$

$$r = \sqrt[4]{\frac{600 \cdot 1.0^3}{2\pi \cdot 10^7 \cdot 0.005}} = 0.209 \text{ in}$$

Feature B

Known: $P = 600 \text{ lbf}$ $l = 1.5 \text{ in}$ $w = 0.75 \text{ in}$ $E = 10^7 \text{ psi}$ $\delta = 0.005 \text{ in}$
 Unknown: Minimum thickness t

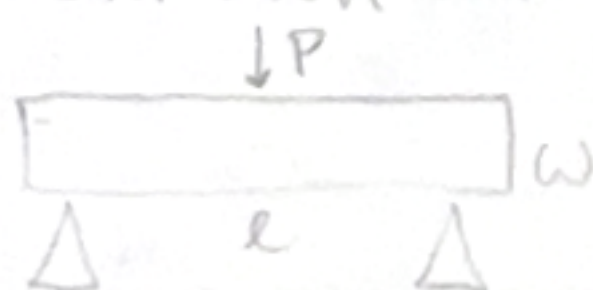


$$\delta = \frac{P \cdot l}{A \cdot E} = \frac{P \cdot l}{w \cdot t \cdot E} \rightarrow t = \frac{P \cdot l}{w \cdot E \cdot \delta}$$

$$t = \frac{600 \cdot 1.5}{0.75 \cdot 10^7 \cdot 0.005} = 0.024 \text{ in}$$

Feature C

Known: $P = 600 \text{ lbf}$ $l = 2.5 \text{ in}$ $w = 1.0 \text{ in}$ $E = 10^7 \text{ psi}$ $\delta = 0.005 \text{ in}$
 Unknown: Min. thickness t

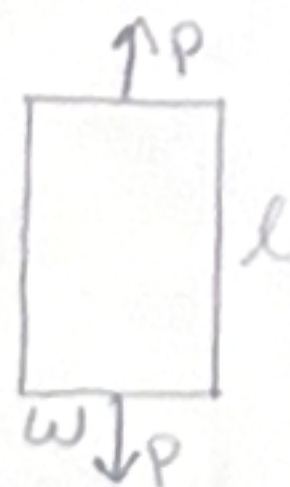


$$\delta = \frac{P \cdot l^3}{48EI} \quad I = \frac{w \cdot t^3}{12} \quad t = \sqrt[3]{\frac{P \cdot l^3}{4wE\delta}}$$

$$t = \sqrt[3]{\frac{600 \cdot 2.5^3}{4 \cdot 1.0 \cdot 10^7 \cdot 0.005}} = 0.36 \text{ in}$$

Feature D

Known: $P_{\text{wall}} = 300 \text{ lbf}$ $l = 1.5 \text{ in}$ $w = 1.0 \text{ in}$ $E = 10^7 \text{ psi}$ $\delta = 0.005 \text{ in}$
 Unknown: Min. thickness t



$$t = \frac{P_{\text{wall}} \cdot l}{w \cdot E \cdot \delta}$$

$$t = \frac{300 \cdot 1.5}{1.0 \cdot 10^7 \cdot 0.005} = 0.009 \text{ in}$$

Feature E

Known: $P_{\text{flange}} = 300 \text{ lbf}$ $l = 1.0 \text{ in}$ $w = 1.0 \text{ in}$ $E = 10^7 \text{ psi}$ $\delta = 0.005 \text{ in}$
 Unknown: Min. thickness t



$$\delta = \frac{P \cdot l^3}{3EI} = \frac{4 \cdot P \cdot l^3}{w \cdot E \cdot t^3}$$

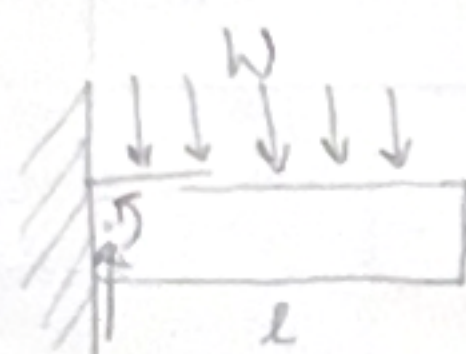
$$t = \sqrt[3]{\frac{4 \cdot P \cdot l^3}{w \cdot E \cdot \delta}}$$

$$t = \sqrt[3]{\frac{4 \cdot 300 \cdot 1.0^3}{1.0 \cdot 10^7 \cdot 0.005}} = 0.288 \text{ in}$$

Feature A

Knowns: $W = 600$ $SF = 4$ $l = 1.0$ in (Assumed) $S_y = 40000$ psi

Unknown: Minimum cylinder radius r



$$\sigma = \frac{Wl}{2Z}$$

$$\sigma = \frac{S_y}{SF}$$

$$Z = \frac{SF \cdot W \cdot l}{2 \cdot S_y}$$

$$r = \sqrt[3]{\frac{4 \cdot Z}{\pi}}$$

$$\sigma = \frac{40000}{4} = 10000 \text{ psi}$$

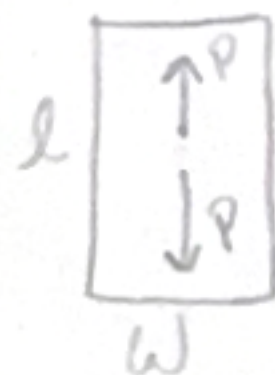
$$Z = \frac{4 \cdot 600 \cdot 1.0}{2 \cdot 40000} = 0.03 \text{ in}^3$$

$$r = \sqrt[3]{\frac{4 \cdot 0.03}{\pi}} = 0.337 \text{ in}$$

Feature B

Knowns: $P = 600$ lbf $l = 1.5$ in $w = 0.75$ in

Unknown: Min. thickness t



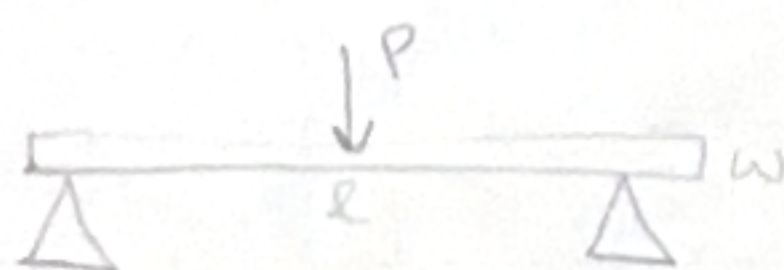
$$\sigma = \frac{P}{A} = \frac{P}{w \cdot t} \quad t = \frac{P}{w \cdot \sigma_{allow}}$$

$$t = \frac{600}{0.75 \cdot 10000} = 0.08 \text{ in}$$

Feature C

Known: $P = 600$ lbf $l = 2.5$ in $w = 1.0$ in

Unknown: Min. thickness t



$$M = \frac{P \cdot l}{4}$$

$$\sigma = \frac{M \cdot c}{I} = \frac{6M}{w \cdot t^2} \quad t = \sqrt{\frac{6M}{w \cdot \sigma_{allow}}}$$

$$M = \frac{600 \cdot 2.5}{4} = 375 \text{ lb-in}$$

$$t = \sqrt{\frac{6 \cdot 375}{1.0 \cdot 10000}} = 0.474 \text{ in}$$

Feature D

Known: $P_{wall} = \frac{600}{2} = 300$ lbf $l = 1.5$ in $w = 1.0$ in

Unknown: Min. thick t



$$\sigma = \frac{P_{wall}}{w \cdot t}$$

$$t = \frac{P_{wall}}{w \cdot \sigma_{allow}}$$

$$t = \frac{300}{1.0 \cdot 10000} = 0.03 \text{ in}$$

Feature E

Known: $P_{hinge} = 300$ lbf $l = 1.0$ in $w = 1.0$ in

Unknown: Min. thickness t



$$M = P_{hinge} \cdot l$$

$$t = \sqrt{\frac{6M}{w \cdot \sigma_{allow}}}$$

$$M = 300 \cdot 1.0 = 300 \text{ lb-in}$$

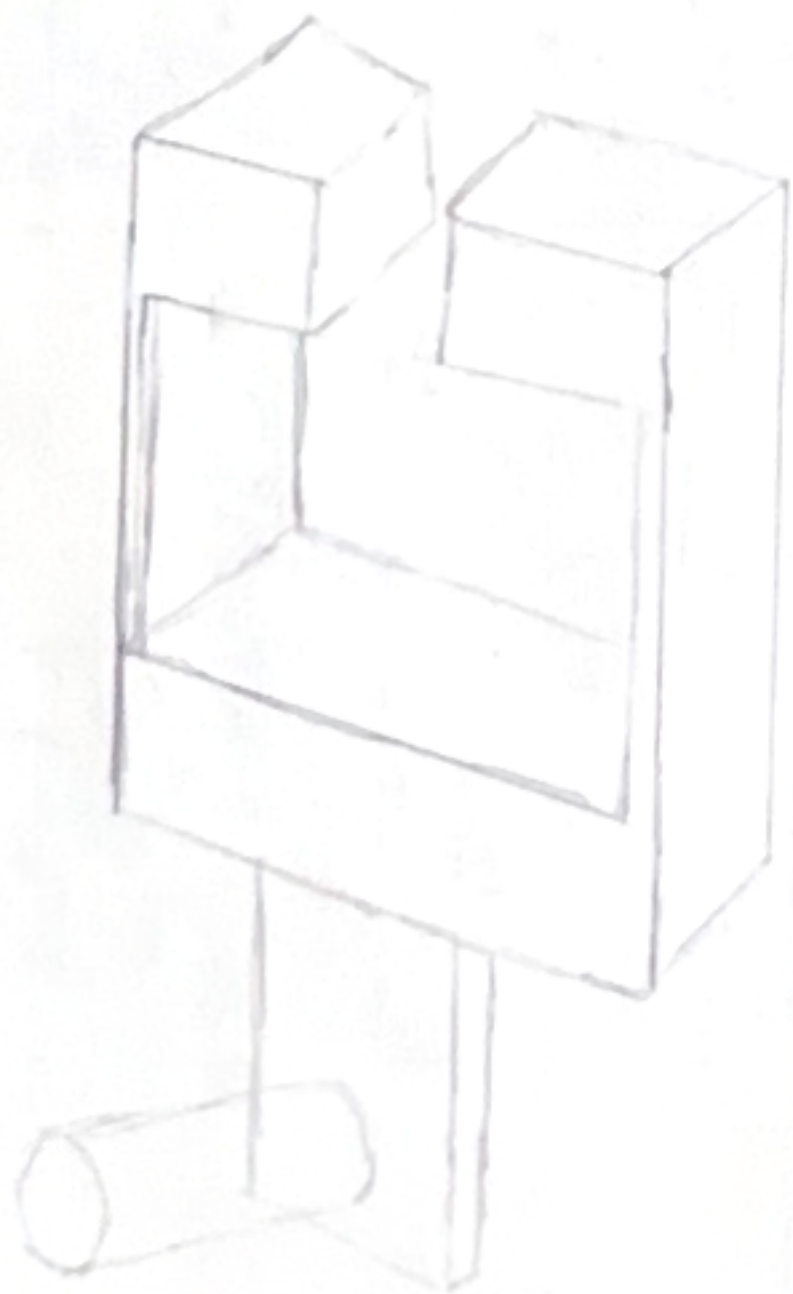
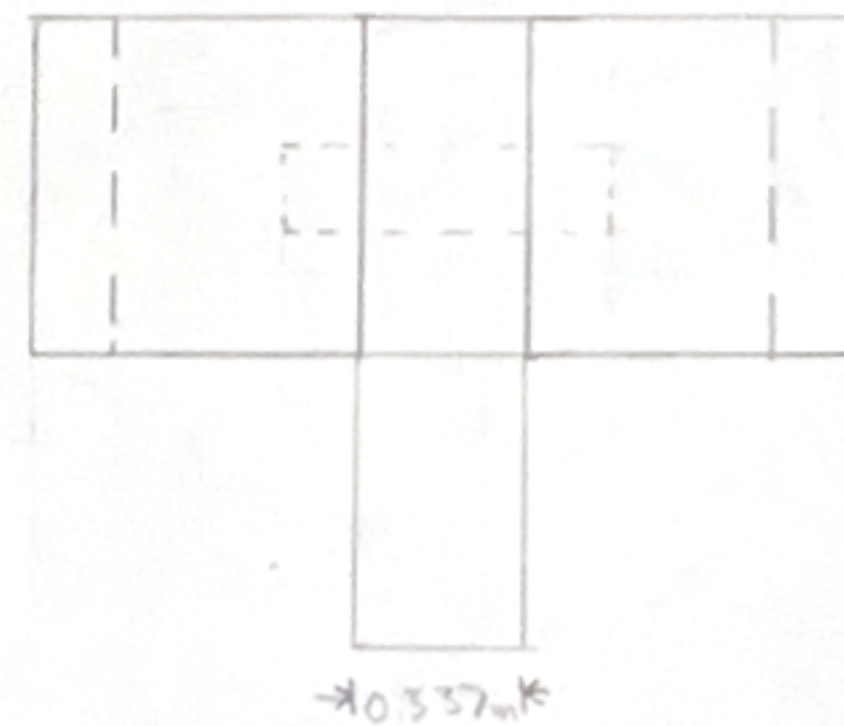
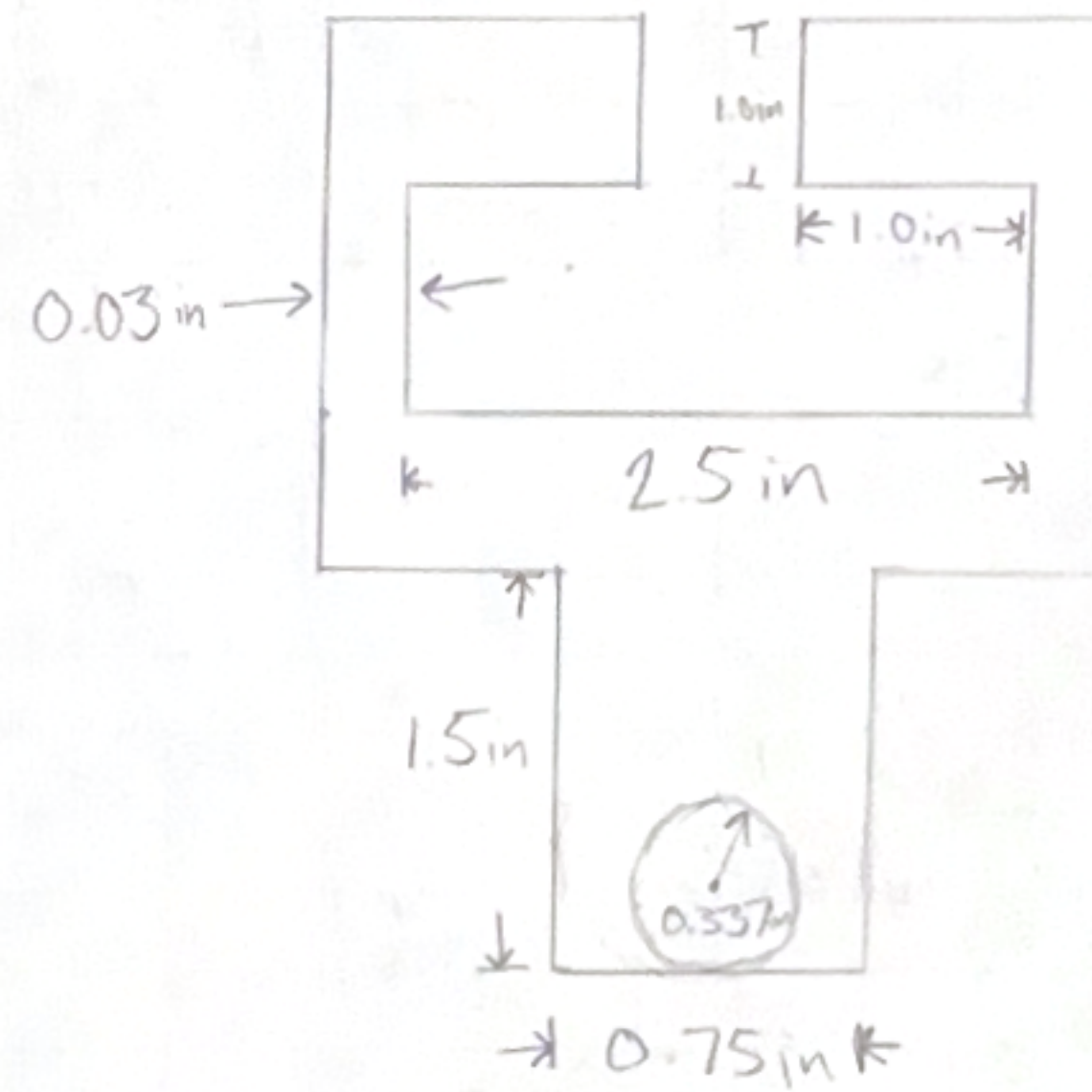
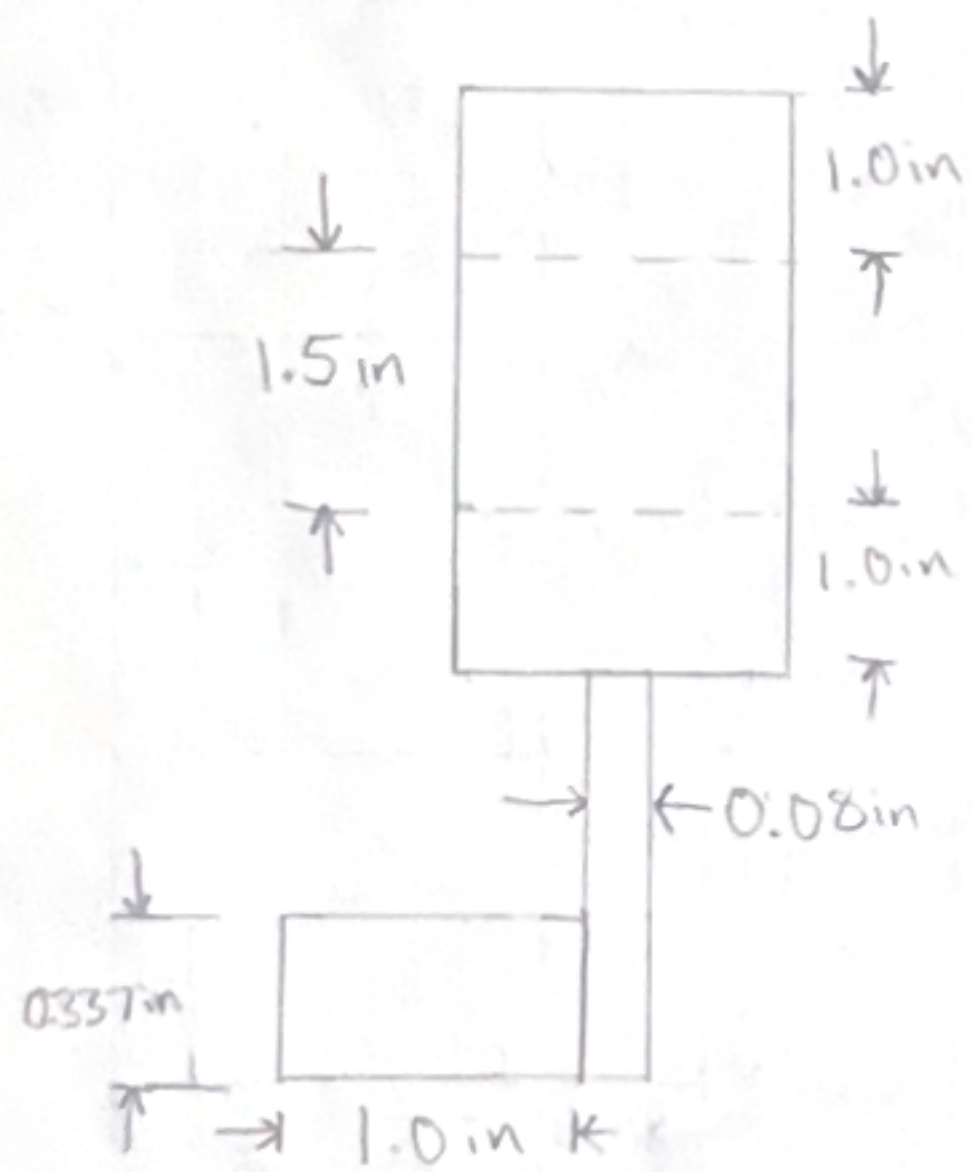
$$t = \sqrt{\frac{6(300)}{1 \cdot 10000}} = 0.424 \text{ in}$$

Stress Analysis Sketch

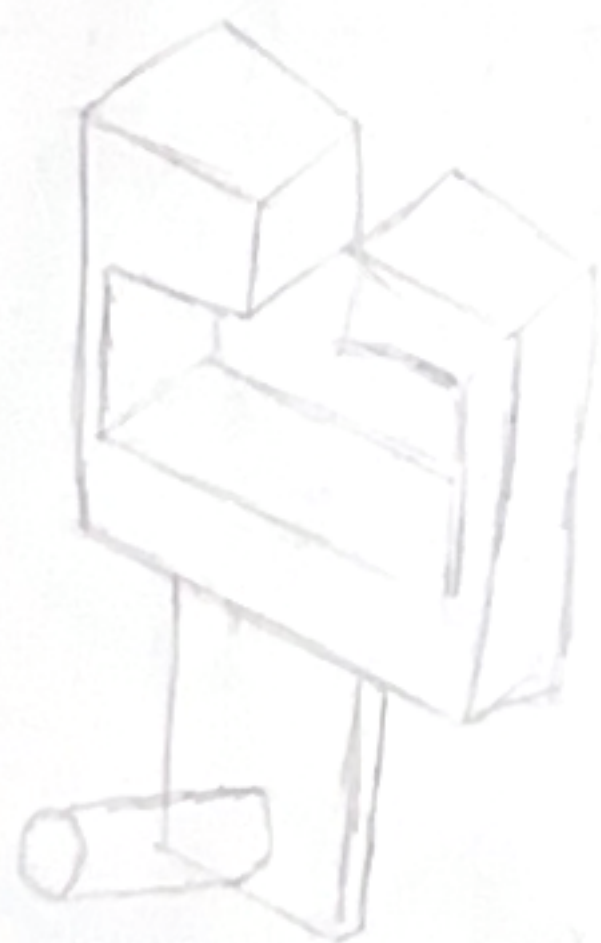
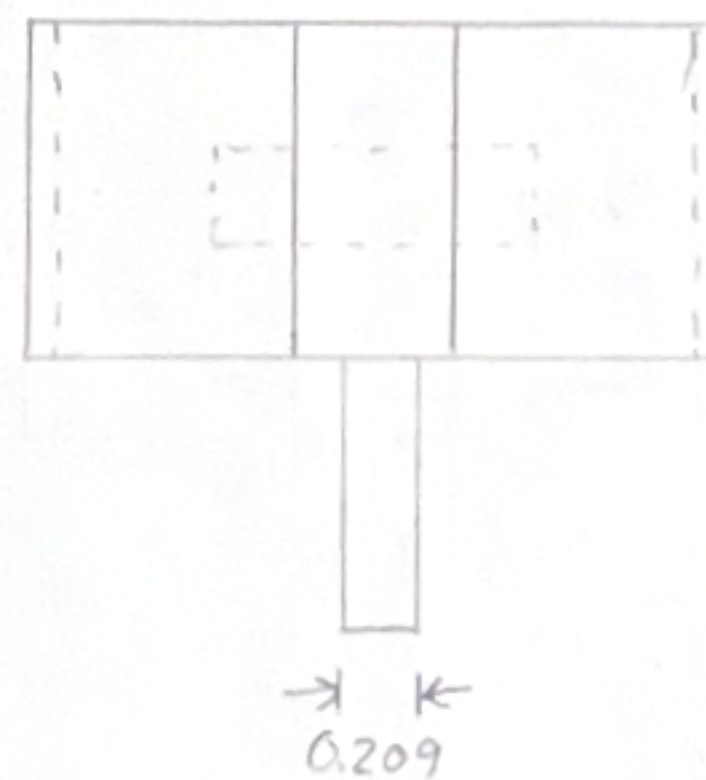
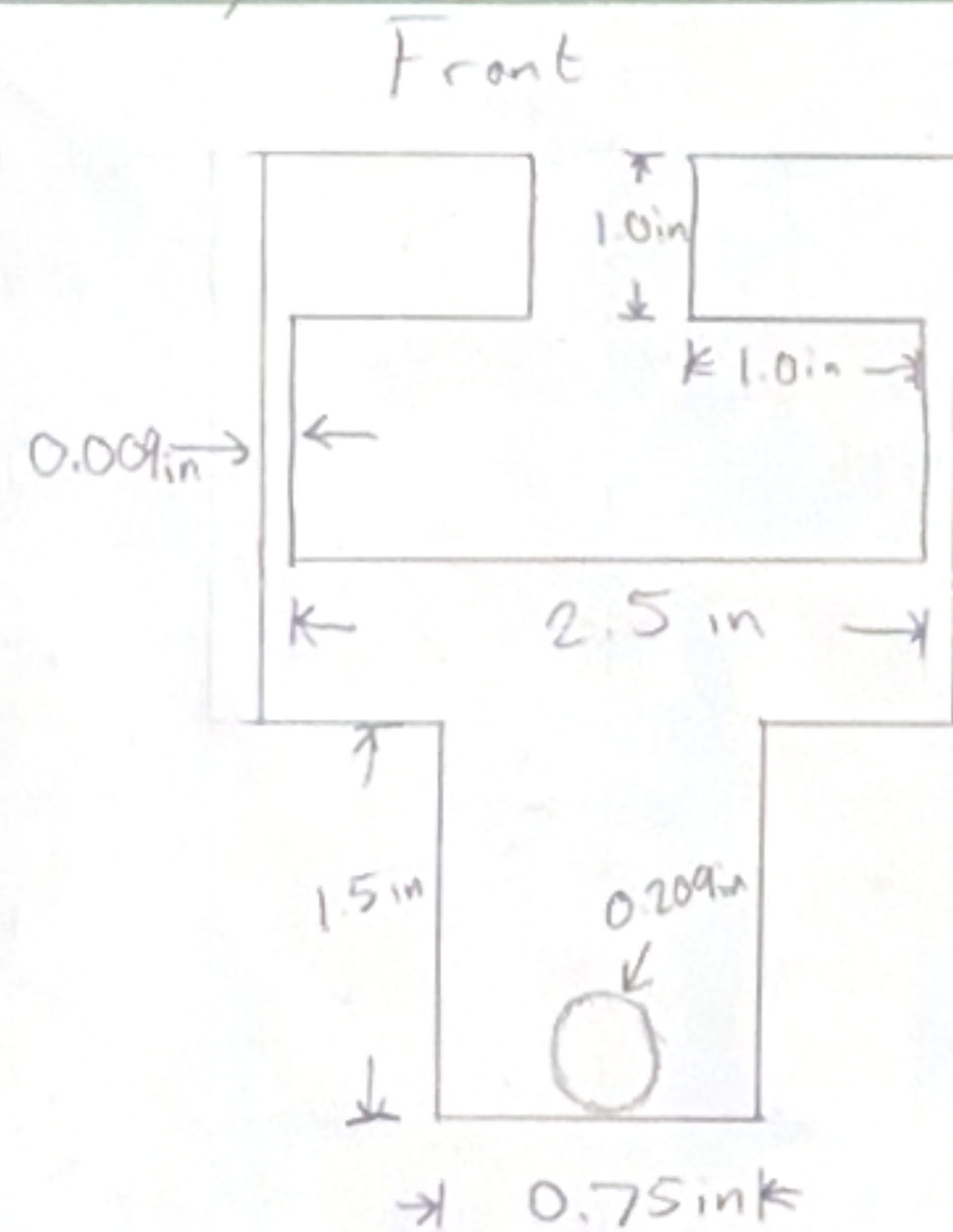
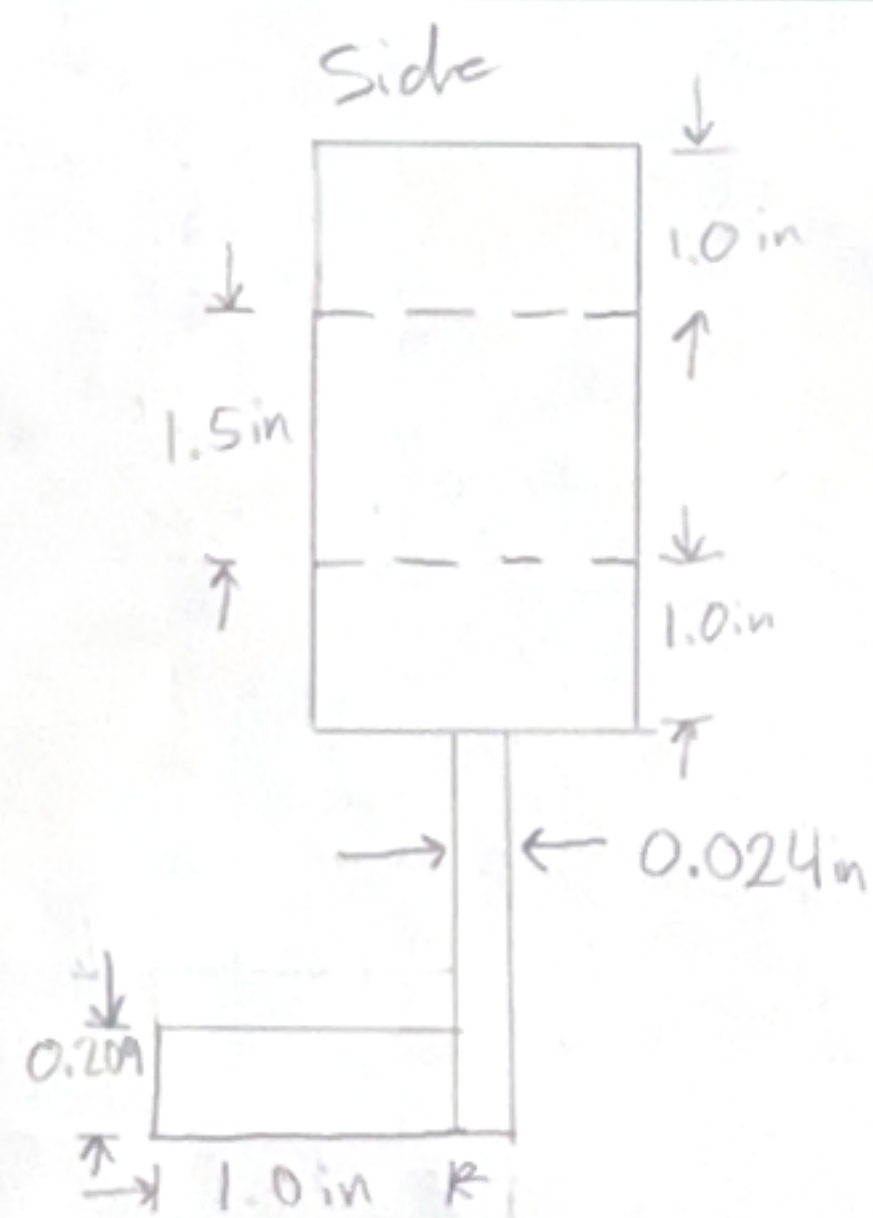
Randy Hwang

15: Bracket Design

Stress Sketch



Stiffness Analysis Sketch



Randy Hoang

AS: Bracket Design Stiffness Sketch